ZERO DAY ATTACK DETECTION AND PREVENTION

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BACHELOR OF TECHNOLOGY

BRANCH: INFORMATION TECHNOLOGY

of Anna University

NOVEMBER 2025

DEPARTMENT OF INFORMATION TECHNOLOGY

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY
BRANCH: INFORMATION TECHNOLOGY
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NOVEMBER 2025

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ACKNOWLEDGEMENT

We express our sincere thanks to **Dr.K.Prakasan,** Principal, PSG College of Technology for providing an opportunity to take up the project work.

We are greatly indebted to **Dr.Umamaheswari K,** Professor and Head, Department of Information Technology for her kind support and encouragement given to complete the project.

We profusely thank our program coordinator **Dr.Karthika Renuka D,** Professor, for constant support and encouragement for the completion of the project.

We express our sincere thanks to our tutor **Dr.Sangeetha B**, Associate Professor, Department of Information Technology, for her constant support and guidance that played a vital role in completing our project.

We express our sincere thanks to our project guide **Ms.Padmashani R,** Assistant Professor (Sl.Gr.), Department of Information Technology, for her constant support and guidance that played a vital role in completing our project.

We would like to extend our sincere gratitude for the invaluable support and collaboration throughout the project. Their guidance, resources, and encouragement have been instrumental in the successful completion of our work.

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ABSTRACT

This project introduces the creation and the actualization of a smart zero-day attack detection and prevention system, which can identify and neutralize in real-time both known and unknown cyber threats. In order to detect deviations from normal network behavior indicative of potential attacks, the system incorporates deep learning models that develop normal network behavior from the network. Deep Autoencoder and Causal Recurrent Network models use only normal traffic for training in order to learn both the structural and temporal features of the data and thus they can capture a complete description of the network activity baseline. The hybrid mechanism, which is the fusion of these models, improves the anomaly detection accuracy because it allows more structural and temporal features to be taken into account for the network analysis. The supervised classifier, which receives detected instances of anomalies, then discriminates between known attack categories and novel incidence patterns by which it can also label zero-day intrusions. To provide real-time interaction with the user or the system architecture, a master–client deployment model has been employed by which client systems are responsible for the continuous network traffic monitoring and only the pre-processed features are sent to the centralized master node for detailed analysis. After detecting abnormal or even malicious behavior, the system, without human intervention, executes preventive measures such as IP blocking or quarantining the source node. The fault-tolerant, accurate, and adaptive cybersecurity framework thus is the result of the seamless integration of anomaly detection, deep learning, fusion-based analysis, and automated intrusion prevention.

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CHAPTER 1

INTRODUCTION

In the fast revolution of digital era, cyberattacks in fact are turning more and more complex and less predictable, containing as a main component the zero-day attacks which can be considered as one of the biggest, most complicated and hardest to fend off challenges in the sphere of network security. These attacks make use of vulnerabilities that are totally unknown or undiscovered in the time window before the release of security patches, thereby they are almost impossible even by traditional signature-based Intrusion Detection Systems (IDS) to detect. The need to have security measures capable of identifying these zero-day exploits and, additionally, of preventing these attacks in real-time has become urgent as a consequence of the constant growth.

As an alternative to the seriously outdated security solutions of already known threats, more and more are the cases where organizations try to solve their issues through intelligent anomaly-based intrusion detection powered by Machine Learning (ML) and Deep Learning (DL) techniques. The presented piece of work is aimed at having the hybrid zero-day attack detection and prevention system that integrates deep learning, attention fusion, and automated intrusion response components designed and put in practice as the first step towards the next eventual solution to the issue.

This model uses a deep autoencoder to identify anomalies, Causal Recurrent Networks (CRN) to analyze the temporal behaviour and a fusion method of errors to merge structural and temporal events for obtaining even greater accuracy. A supervised classifier (CatBoost) is used to identify known attacks, while Centroid Distance Thresholding is utilized for detecting new intrusions that may thus be considered zero-day attacks. The architecture of this system is a master–client one where the client nodes that may be the different points of the network, capture the data concerning the traffic and then send it to the central, master node, which performs the detection and consequently triggers the prevention actions in real-time through an automatic IP blocking mechanism.

1.1 Problem Statement

Traditionally, IDS are mainly based on predefine signatures or known attack patterns to identify malicious activities. As a result, they cannot effectively detect zero-day attacks that leverage unrecognized vulnerabilities or create new ways of attacks.

The extreme unpredictability and the very fast changes of zero-day attacks are the main reasons why there is such a huge gap in network defenses between the attackers and their victims. Consequently, attackers can infiltrate systems and remain undetected by traditional IDS.

Therefore, the demand for an intelligent anomaly-based system capable of recognizing unknown network activities in real-time, being able to evolve with the threats and taking preventive measures if only a few can be harmed, has increased enormously.

1.2 Objectives

- To create an anomaly-based hybrid intrusion detection and prevention system that automatically detects known attacks and zero-day attacks in real time.
- The purpose is to devise a Deep Autoencoder (DAE) model and CRN which understand the usual network traffic and recognize any deviations as anomalies.
- By integrating fusion mechanisms that combine multiple data views (structural and temporal events) to achieve higher anomaly sensitivity.
- Implementing a supervised classification model that identifies known attack categories and flags unknown threats as zero-day intrusions by means of Centroid Distance Thresholding.
- Constructing a real-time Intrusion Prevention System (IPS) that can henceforth halt a suspicious IP or isolate it in case of a high anomaly score.
- The intent is to employ the framework through a master–client architecture for centralized analysis and distributed data monitoring at several nodes.

1.3 Scope of the Project

- Experimenting with DAE and CRN - based models for anomaly detection without the need for supervision.
- Producing the fusion framework which merges the various models' output.
- Developing a supervised classifier with Centroid Distance Thresholding to dynamically differentiate between known and unknown attacks.
- Creating a real-time IPS capable of automatic IP blocking upon a detection implementation.
- Setting up a master–client architecture that supports distributed data monitoring and centralized control.
- Employing different tools such as Wireshark for traffic capture, TensorFlow/PyTorch for model training, and Flask for real-time communication between nodes.

1.4 Motivation

Zero-day vulnerabilities are the most desirable aims of cybercriminals and Advanced persistent threats (APT) because these vulnerabilities give them a chance to exploit a system before an official patch is issued. The recent cybersecurity reports indicate that there is a huge increase in the number of these types of incidents, as well as in their severity. Normal IDSs that are only effective against already known threats, are not strong enough to confront such newly originated and continuously changing attacks. Such a situation necessitates the creation of an anomaly-based, self-learning and thus indefinitely adaptable framework that can recognize even the newest types of malicious behaviour. The target of the introduced solution is to marry the capabilities of DL-based anomaly detection with the automation of intrusion prevention in order to close the security gap that exists between detection and intervention.

1.5 Real-World Importance

- Enhanced Cyber Defense: Detects complex attack scenarios that a signature-based IDS might not be able to identify.
- Adaptive Security: Keeps getting better as it learns from newer traffic data without the need for any manual intervention.
- Scalable Deployment: Can be effectively used in large network environments, cloud.
- Proactive Response: Without any human intervention, it lessens the effects of the eventualities that it identifies by e.g. stopping or severing interaction with the suspicious source thereby, reducing the time to respond and the risk of further damage.

1.6 Cryptographic Enhancements

- Deep Autoencoder-Based Detection: Understands normal traffic patterns and considers those instances as anomalies, which have been reconstructed with high error, i.e., it flags such instances.
- Temporal Modeling with CRN: Uses LSTM layers to capture the sequential nature of traffic patterns and identify anomalies that are dependent on the given context.
- Fusion: Combines structural and temporal events to uplift the model accuracy and stability.
- Real-Time Intrusion Prevention: In doing so, it can be said that the actions most definitely taken are those of an intruder such as automatically blocking malicious IP addresses or, e.g., command-line or firewall-based rules can be used for isolating the affected nodes.
- Centralized Monitoring (Master–Client Model): Allows the collection of data from different locations to be done in a decentralized manner while the decision-making is done from a central point, thus making it scalable.

1.7 Challenges & Considerations

- Data Imbalance: The datasets from the real world usually have few attack examples than normal traffic, which affects the balance of the training.
- Model Generalization: The main issue is how to keep the model working at the same level in different network environments and with new attack types.

- Real-Time Processing: The main concern is how the detection and prevention can be done with a very short time interval in large-scale networks.
- Scalability: The issue is how to deal with numerous client nodes and at the same time ensure that the data are transmitted securely and efficiently.

1.8 Significance of Study

This project advances the intelligent cybersecurity and network defense domain with the main emphasis on:

- Making the intrusion detection systems more powerful by adding deep learning-based anomaly detection that can detect both known and zero-day attacks in real time.
- Using fusion mechanisms that merge structural and temporal features for getting better detection accuracy and fewer false positives.
- Creating an automated intrusion prevention module that can block or isolate the malicious traffic sources without the need of a human.
- Developing a scalable master–client architecture for distributed monitoring, which allows the efficient deployment, enterprises, and IoT environments.

This initiative, through the integration of anomaly detection, classification, and prevention in a single framework, is thus offering a strong, flexible, and scalable shield against the fast-changing zero-day threats, which is very responsive and efficient from an operational point of view.

CHAPTER 2
LITERATURE SURVEY

2.1 Literature Review

This chapter reviews existing research on zero-day attack detection and prevention using Machine Learning (ML), Deep Learning (DL). Traditional Intrusion Detection Systems (IDS) depend heavily on predefined attack signatures, which makes them ineffective in identifying zero-day attacks that exploit previously unknown vulnerabilities. With the rapid evolution of cyber threats, there is an increasing need for intelligent, adaptive, and data-driven security models that can detect anomalies in real-time without relying on prior attack definitions.

Recent advancements in intrusion detection research have focused on unsupervised and hybrid learning techniques that analyse behavioural patterns in network traffic to detect potential zero-day attacks. Deep Autoencoders, Causal Recurrent Networks (CRNs), and ensemble models have emerged as promising solutions due to their ability to learn complex traffic representations and identify deviations from normal behaviour.

In addition, fusion mechanisms have been adopted to integrate multiple feature spaces — such as structural and temporal data — to enhance robustness and improve detection performance. Studies have also explored transfer learning and Zero Shot Learning (ZSL) techniques to identify unseen attack types by transferring knowledge from known attack categories. While these techniques enhance the ability to generalize across unseen patterns, their dependence on semantic mapping and pre-learned data distributions reduces their performance against highly novel zero-day threats.

Overall, existing literature highlights a continuous trade-off between detection accuracy, interpretability, and real-time performance. Although several approaches have achieved significant progress in identifying unknown or evolving threats, most existing systems remain limited by their computational requirements and lack of real-world adaptability. To address these shortcomings, the proposed project introduces a hybrid deep learning-based intrusion detection and prevention framework that integrates Deep Autoencoders for feature reconstruction, LSTM-based Contextual Recurrent Networks (CRN) for temporal modeling, and fusion for dynamic anomaly scoring. Furthermore, the classifier enables distinction between known and zero-day attacks.

The study by J. Smith, L. Brown, and A. George, “A Combined Classification Approach for Zero-Day Attack Detection in Computer Networks,” proposed an ensemble-based hybrid classification method for identifying zero-day attacks. By leveraging multiple machine learning classifiers within an ensemble structure, the authors demonstrated that detection accuracy can be significantly improved compared to traditional single-model approaches. However, they also highlighted that the computational complexity of ensemble learning introduces latency issues, limiting its practical use in real-time, high-speed network environments.

M. Sarhan, K. Layeghy, and M. Portmann, “Zero-Shot Learning for Unknown Intrusion Detection,” explored the application of ZSL to identify previously unseen intrusions by transferring knowledge from known attack types. Their model successfully detected novel attack behaviors without requiring prior signatures or retraining, showing promising generalization to new threats. Nonetheless, the authors observed that model performance declined when tested on entirely unrelated attack scenarios, revealing a limitation in the model’s adaptability.

A. Kumar, D. Shah, and N. Reddy, “Unsupervised Anomaly Detection Using Clustering and Autoencoders,” introduced an unsupervised framework combining clustering algorithms with deep autoencoders to detect anomalies in network traffic without relying on labeled datasets. This approach suffered from a higher false-alarm rate due to the absence of supervision during training, which reduced overall detection stability.

R. Patel and T. Zhang, “CNN-Based Anomaly Detection Model with Regularization for IoT Network Security,” proposed a convolutional neural network enhanced with L1 and L2 regularization to detect zero-day attacks in IoT environments. The model outperformed conventional machine learning classifiers and achieved high detection accuracy. However, the authors noted that the computational requirements of CNNs make them less suitable for real-time use on lightweight IoT devices with limited hardware resources.

L. Thomas, J. Fernando, and P. Roy, “A Sandbox-Based Machine Learning Framework for Malware Detection,” presented a sandbox-driven system that utilizes machine learning models to analyze behavioral features extracted from controlled sandbox environments. This technique improved the detection of malicious software and zero-day threats but was found to be resource-intensive, rendering it inefficient for large-scale purpose.

S. Banerjee, H. Wong, and V. Mehta, “Semi-Supervised Learning Models for Zero-Day Intrusion Detection,” proposed a semi-supervised approach combining clustering and classification to detect unknown intrusions using minimal labeled data. Their results indicated that semi-supervised learning improves adaptability to evolving attack patterns, though noisy traffic conditions increased false-positive rates and impacted system dependability.

B. Dahal, K. Bajgai, and R. Rahimi, “Integration of Explainable Artificial Intelligence with MLP for Zero-Day Attack Detection,” developed a framework that merges Explainable Artificial Intelligence (XAI) with Multilayer Perceptron (MLP) networks to improve interpretability in decision-making. The integration enhanced transparency and trust in model predictions but showed scalability limitations and slower training times when applied to large datasets.

In a subsequent work, B. Dahal, L. Nguyen, and A. Verma, “Attention-Based Explainable AI Fusion for Intrusion Detection in IoT Networks,” extended this concept by integrating attention mechanisms with XAI to improve both interpretability and precision in zero-day detection. The model exhibited higher accuracy and better feature-level insights but remained computationally heavy, restricting its deployment in real-time IoT environments.

Y. Lee, T. Choi, and M. Hassan, “A Hybrid Ensemble Framework for Unseen Intrusion Detection,” proposed combining ensemble classifiers with unsupervised algorithms such as Isolation Forest and clustering to detect previously unseen intrusions. The hybrid approach improved system robustness and reduced bias toward known attack classes. However, the authors reported persistent false positive challenges and a lack of detailed comparative evaluation across multiple datasets.

Finally, Y. Guo, “A Review on Machine Learning-Based Zero-Day Attack Detection Techniques,” conducted an extensive review of modern zero-day detection methods using machine learning and deep learning. The study emphasized the growing need for adaptive hybrid systems capable of balancing accuracy, scalability, and interpretability while minimizing computational overhead. Guo concluded that future IDS research should focus on real-time adaptability and energy-efficient deployment without compromising detection precision.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Project Specification

The essential elements of the proposed Hybrid Zero-Day Attack Detection and Prevention System are the DL-based anomaly detection and the automated prevention system to enhance the network defense mechanisms. It utilizes Deep Autoencoder (DAE) for unsupervised anomaly detection, Causal Recurrent Network (CRN) for sequential pattern analysis, and a fusion mechanism that combines structural and temporal error scores to have a higher level of detection accuracy and robustness.

A supervised classifier is in charge of identifying known attacks, and Centroid Distance Thresholding is responsible for finding the previously unseen patterns as potential zero-day attacks. The system is implemented through a master-client architecture, where distributed client nodes perform network traffic data capturing and transmitting to a centralized master server for analysis and automated IP blocking in case of anomaly detection. Thus, this architecture assures real-time responsiveness, scalability, strong intrusion prevention in enterprise and IoT environments.

3.2 Functional Requirements

- Data Acquisition and Preprocessing: It obtains live network data through the use of tools like Wireshark, followed by feature extraction, normalization, and encoding to make the data compatible with the model.
- Deep Autoencoder (DAE) for Structural Anomaly Detection: The model ingrains typical traffic patterns and, through reconstructing the input, determines the parts of the data that differ from common patterns, and these parts are thus considered anomalies.
- Causal Recurrent Network (CRN) for Temporal Analysis: The use of time-dependent traffic sequences enables the network to detect changes in flow that could not be detected by a static model, thus the network can recognize anomalies in the time series.
- Fusion Mechanism for Hybrid Anomaly Detection: The system merges errors from reconstruction and prediction with the help of a weighted attention fusion strategy in order to accomplish a higher detection capacity as well as a lower false alarm rate.

- Supervised Classification (CatBoost): Through the use of labeled datasets and confidence thresholds, the system segregates the known threats from the unknown activities while classifying the known ones.
- Zero-Day Detection through Centroid Distance Thresholding: Identifying new or very few instances of data as a possible zero-day attack by measuring the distance from the centroid of the known class clusters is the method used here.
- Automated Intrusion Prevention: On the detection of the offending event, the system through a dynamic firewall or command-line rules performs the real-time blocking or isolation of the suspicious IPs.
- Master–Client Communication: RESTful APIs powered by Flask or Socket.IO allow for non-blocking message transmission between distributed nodes and the central detection server.

Metrics of the Functional Requirements

- Reconstruction Error (Mean Squared Error): A measure of how well the DAE can reconstruct typical network patterns.
- Prediction Deviation (L2 Distance): Represents the temporal prediction of the CRN and pinpoints behavioral changes.
- Fusion Accuracy: Reflects the overall performance of the anomaly detection after the fusion of DAE and CRN errors.
- Classification Accuracy (Precision, Recall, F1-Score): The CatBoost model's capability to correctly identify known attacks is measured by these metrics.
- Zero-Day Detection Rate: Indicates the fraction of new threats that have been correctly detected by centroid thresholding.
- Response Time for IP Blocking: The interval between detection and the automatic isolation of an IP, which is recorded as latency.

3.3 Non-Functional Requirements

- Performance: The system is capable of handling traffic data that is continuously coming in, in a very short time (real-time) with a very low reaction time (latency) in the detection and prevention operations.

- Accuracy and Reliability: The system keeps very high detection accuracy even when it is loaded with different network traffic and is stable in operation under different network conditions.
- Scalability: The system is able to handle the situation when several client nodes are sending traffic data at the same time to a central detection unit without losing the performance.
- Extensibility: The system architecture is very flexible and thus it allows you to add new machine learning or deep learning models without needing to change the whole system.
- Availability: The system is able to maintain a continuous uptime of at least 99% and thus it ensures network monitoring and protection without any interruptions.

Metrics of the Non-Functional Requirements

- Detection Latency: The time duration that is measured starting from the packet capture until the output of anomaly detection.
- Model Accuracy Metrics (Precision, Recall, F1-Score): Represents a set of metrics that measure the model's ability to detect known and unknown attacks correctly.
- Scalability Index: Performance evaluation based on throughput and latency changes with an increasing number of clients.
- Resource Utilization (CPU/Memory Usage): The measure of the system's computational efficiency during model inference.
- System Uptime Percentage: The time that is used to measure the system's trustworthiness and continuous availability for uninterrupted operation.

3.4 System Requirements

The system requires the following hardware and software components:

Software Requirements:

- Operating System: Windows 10 or later
- Programming Language: Python 3.8 or above
- Libraries/Frameworks:

i. TensorFlow 2.x or PyTorch (for Deep Autoencoder)

ii. Scikit-learn (for preprocessing, evaluation)

iii. Pandas, NumPy (for data manipulation)

iv. Google Colab (for development/testing)

- Traffic Monitoring Tools: Wireshark (for packet capture on clients)

Hardware Requirements are needed as shown in Table 3.1.

Table 3.1 Hardware Requirements

System Requirements	Master System	Client Systems
Processor	Intel Core i5/i7	Intel Core i3 or equivalent
RAM	Minimum 8 GB (16 GB recommended for real-time processing)	4 GB minimum
Storage	SSD with at least 50 GB free	20 GB
Network	Ethernet/Wi-Fi (LAN recommended for faster log transfer)	Basic connectivity to send logs to master

3.5 Network Requirements

Although the system primarily operates locally or in cloud-based notebook environments, certain network requirements are essential for data access, collaboration, and performance monitoring.

- **Stable Internet Connection:**
  Required for downloading datasets (e.g., from Kaggle) and saving trained models or logs to cloud storage.
- **Cloud Compatibility:**
  The project should support platforms like Google Colab and Kaggle, which allow GPU acceleration and shared environment access.

- **Data Transfer Efficiency:**
  Compressed data formats and caching mechanisms should be used to reduce load times for large datasets.
- **Collaboration and Versioning:**
  Code and results should be version-controlled through GitHub for team-based development and reproducibility.
- **Secure Data Handling:**
  All datasets used must remain anonymized and handled securely, ensuring no patient identity exposure during transmission or processing.
- **Visualization and Monitoring:**
  Remote logging tools like TensorBoard or WandB can be optionally integrated for real-time performance tracking during training.

  By fulfilling these network and system requirements, the framework ensures reliable training, smooth deployment, and accessibility for further research extensions.

CHAPTER 4

PROPOSED METHODOLOGY

The Zero-Day Attack Detection and Prevention Proposed methodology is intended to build a smart, self-changing, and expandable security system that can learn, identify, and solve security problems in the network at the same time. Compared to regular Intrusion Detection Systems (IDS) that are mostly dependent on static rule sets and predefined signatures, this solution adopts a data-driven, behavior-based viewpoint.

The aim is to instill the idea deeply into the system of the collocation of three complementary learning parts: a Deep Autoencoder (DAE) for structural anomaly detection, a Causal Recurrent Network (CRN) for temporal sequence modeling, and a CatBoost classifier for supervised attack categorization. The CatBoost classifier plays the role of supervised attack analysis when the first-step detection has been done by the fusion of reconstruction and prediction errors. CatBoost (Categorical Boosting) is a very powerful gradient-boosted decision tree algorithm that has been optimized for dealing with heterogeneous network data, made up of both categorical and continuous features. It also makes use of Centroid Distance Thresholding, where the data points that fall outside the learned decision boundaries can be regarded as zero-day attacks.

This method's central innovation is in the fusion mechanism that links the outputs of the DAE and the CRN. Rather than depending on the separate anomaly scores, the system uses a weighted fusion approach that merges reconstruction errors (structural anomalies) and prediction deviations (temporal anomalies) into one, integrated anomaly score.

To make the deployment possible in complex and distributed settings, the system as a whole functions under a Client–Master layout. Initial tasks such as packet feature extraction, normalization, and encoding are done by each client, and after that, the processed data is sent to the master node securely. With the assistance of the DAE–CRN–CatBoost pipeline, the master node completes deep inference, fusion, and classification; and if there is any malicious behavior, it can initiate automated prevention mechanisms such as IP blocking, connection termination, or network isolation.

4.1 System Architecture

The overall system consists of four main modules—Deep Autoencoder, Causal Recurrent Network, CatBoost Classifier, and a Client–Master Communication Layer.

- **Deep Autoencoder (DAE):** This module learns compact, latent representations of normal network traffic patterns and identifies structural anomalies by measuring reconstruction errors. It captures hidden correlations among features, enabling the detection of subtle deviations that often signify malicious behavior.
- **Causal Recurrent Network (CRN):** The CRN module models temporal dependencies across consecutive traffic flows and predicts future network behavior to detect time-based anomalies. By understanding causal relationships in packet sequences, it identifies abnormal transitions or sudden shifts in network dynamics that static models fail to capture.
- **CatBoost Classifier:** This component performs supervised classification to distinguish between known attack categories and unknown zero-day activities. Leveraging gradient boosting over decision trees, CatBoost efficiently handles heterogeneous network data, reduces overfitting, and provides explainable feature importance metrics. It also incorporates centroid distance analysis to detect outliers, enabling the identification of novel attack instances with improved accuracy and interpretability.
- **Client–Master Setup:** Distributed client nodes continuously capture local network traffic, perform feature extraction and preprocessing, and securely transmit summarized features to a centralized master node. The master node executes the DAE–CRN–CatBoost pipeline, performs fusion-based anomaly detection, and initiates automated prevention measures such as IP blocking or source isolation.

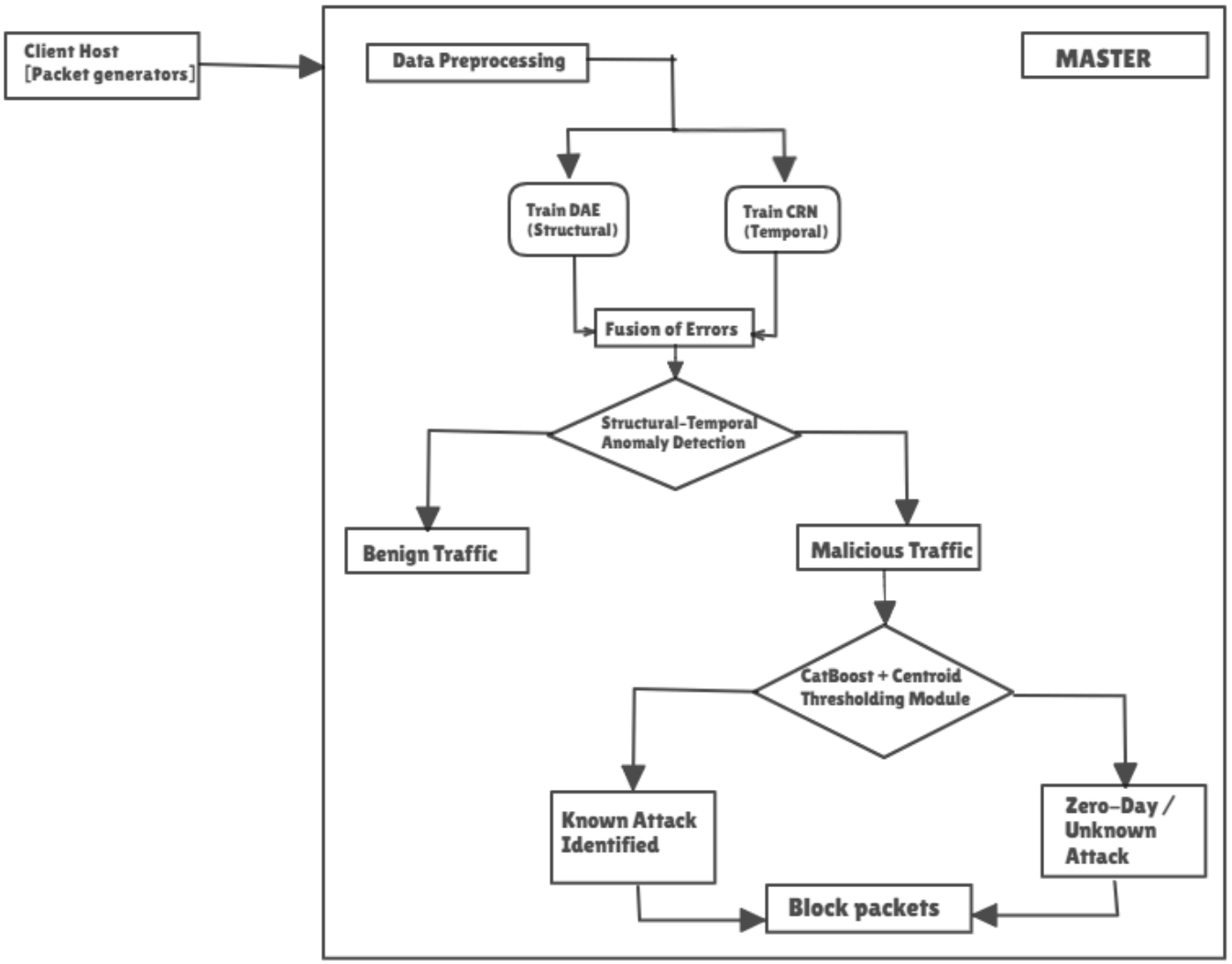


Fig 4.1 Block diagram of the proposed Zero-Day Detection and Prevention framework

In this system as shown in Fig 4.1, the Master Node centralizes data packets from client hosts for Data Preprocessing, including feature extraction and normalization. The core detection uses a Deep Autoencoder (DAE) for structural and a Causal Recurrent Network (CRN) for temporal anomaly analysis. Their outputs are combined via a Fusion of Errors to categorize traffic as Benign or Malicious. Malicious packets are further analyzed using CatBoost + Centroid Thresholding to distinguish Known from Zero-Day (Unknown) Attacks. The final step is an automatic Block Packets response to prevent intrusion by isolating or blocking the malicious sources.

4.2 Dataset Description

The methodology implements a hybrid data collection strategy that enables the model to be robust against zero-day threats in the real world. The approach incorporates both structured public resources and live network telemetry. The UGRansome dataset is a public ransomware and network intrusion dataset explicitly created to facilitate research in anomaly detection, ransomware classification, and zero-day attack identification. It results from the integration of network traffic and artificially generated ransomware communication patterns and was collected and curated during 2018 and 2019. There are about 1,49,044 instances (rows) in the dataset, each one depicting a different example of network behavior or ransomware activity. These examples are detailed using 14 features (columns) that not only describe the low-level network characteristics but also the higher-level behavioral indicators of malware activity.

Dataset primary columns are Time, Protocol, Flag, Family, Clusters, SeedAddress, ExpAddress, BTC, USD, Netflow_Bytes, IPaddress, Threats, Port, and Prediction. These features denote temporal information, network protocol metadata, Bitcoin wallet information (used in ransomware payment tracing), and categorical fields that indicate known or suspected ransomware families. By integrating conventional network flow attributes with ransomware-specific fields, UGRansome becomes a highly potent resource for sophisticated behavioral modeling.

The data describe various kinds of attacks such as Denial of Service (DoS), Port Scanning, SSH Brute Force, Spam, Blacklisted IP activity, and UDP Scan, along with normal and zero-day or unknown attack behaviors. Each record is linked to a predicted label and a closest known class, thus enabling models to distinguish between known attack signatures and possible zero-day behaviors by means of distance metrics or feature deviations.

The dataset's class labels are roughly categorized into three groups:

- Signature (S) – Illustrating the most straightforward and confirmed attack pattern.
- Synthetic Signature (SS) – Created by merging or altering the existing signatures to imitate the new attack behaviors.
- Anomaly (A) – Depicting unknown or zero-day ransomware families whose behavior is not previously observable from the training data.

Therefore, the UGRansome dataset offers a substantial and varied set of network events for the assessment of cybersecurity models. Its combination of fully labeled and semi-labeled samples render it perfect for machine learning-based intrusion detection, behavioral modeling, and zero-day attack classification. The dataset is commonly employed to validate models that depend on distance-based classification, deep autoencoders, clustering, and hybrid ML-DL strategies. Here the goal is to separate known threats from emerging ransomware patterns in real-time network environments.

To get a current and realistic baseline of normal behavior, real-time network traffic was captured continuously. Wireshark/TShark was running on distributed client nodes that were simulating a production environment. This important step involved passive packet monitoring and then flow aggregation to change raw packet streams into consolidated flow records.The real-time captured traffic was carefully filtered and checked to ensure that it only contained benign, non-malicious activity. Thus, it became an exclusive training set for the unsupervised anomaly detection models (DAE and CRN). At the same time, the existing, labeled attack instances from the UGRansome structure, that include different categories such as DoS and Botnet, were separated to train the supervised CatBoost Classifier. Such a dual collection strategy guarantees that the system learns a clean baseline of 'Normal' behavior from live data and at the same time keeps the necessary knowledge of 'Known Attack' signatures from the structured corpus.The finally gathered data is divided into different sets for training, validation, and testing, thus enabling a thorough and comprehensive model evaluation process.

4.3 Data Preprocessing

The different data collected have to undergo a rigorous data preprocessing that is a mandatory step before they can be transformed into an optimal format for the deep learning pipeline in order to maximize the models' efficiency and accuracy. Feature mapping is the very first critical task that ensures all real-time and structured dataset features are strictly in accordance with the project's 14-feature schema. Next comes the handling of categorical features when non-numeric attributes (such as protocol identifiers) are changed with One-Hot Encoding in order to prevent models from wrongly interpreting them as ordinal data. After that, all continuous numerical features are fundamentally scaled and normalized through the use of various methods like Min-Max normalization or standardization. This is done to stop the features with a large range of values from dominating the model weights during training.

In the case of the Causal Recurrent Network (CRN), the sequential nature of network activity is a must which means the normalized flow records have to be changed back into fixed-length temporal sequences or 3D tensors. Minor anomaly suppression is additionally applied to the 'Normal' training subset to obtain the purest baseline for the DAE. The 'Attack' data are manipulated to obtain cluster centroids in the feature space which is a necessary input for the Centroid Distance Thresholding used in zero-day identification. Lastly, the whole preprocessing pipeline is constructed to be a lightweight and distributable pipeline so that it can be run efficiently on the client nodes, thus reducing the data volume and computational load on the central master server. In essence, this is our "data cleaning" step, where we sharpen the signal and discard the noise. This meticulous preparation is vital, ensuring our models don't waste time on raw data clutter but focus purely on the patterns of threat behavior. A standardized input guarantees that the DAE's baseline is rock-solid and impervious to data inconsistencies. This upfront investment in quality directly translates to lower false alarms and faster response times in a critical security scenario. Ultimately, a perfectly preprocessed dataset is our first line of defense against highly evasive zero-day exploits.

4.4 DAE Training Process

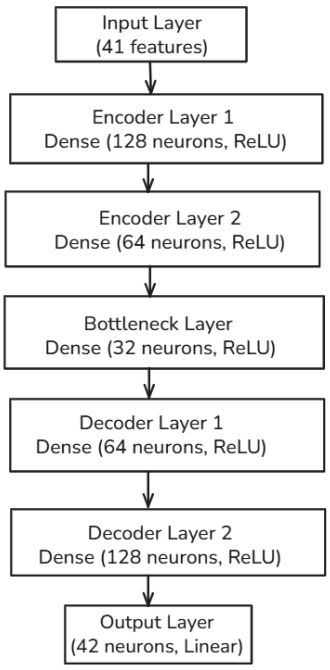


Fig 4.2 DAE Workflow

The Deep Autoencoder model architecture was the main idea to the problem of unsupervised feature learning as well as anomaly detection. As shown in Fig 4.2, the input layer depicts the features present in the dataset, the model through multiple encoding layers compresses the input data into a low-dimensional latent representation and then the symmetric decoding layers reconstruct it again to the original form. The encoder has three Dense layers of 128, 64, and 32 neurons respectively. Each neuron is activated by the ReLU function which helps the model to capture nonlinear dependencies and to thus carry out dimensionality reduction effectively.

The bottleneck layer with 32 neurons stands for the compact latent feature space. The decoder is the same as the encoder with Dense layers of 64 and 128 neurons followed by an output layer activated by a linear function to reconstruct the original input features.

Once the architecture has been set up the model is compiled with the Adam optimizer that handles the network weights during the training in an efficient way. The mean squared error (MSE) loss function is the one used to minimize the reconstruction error, thus it is good for anomaly detection tasks. To avoid overfitting early stopping is set to monitor the loss and get back the best model weights. The model is trained on scaled input data for 100 epochs with a batch size of 256 and an 80–20 train-validation split. At the same time, training and validation loss curves are watched to check if the model is converging. In the end, to training, reconstruction errors are calculated for all data instances and an anomaly threshold is set as the average reconstruction error plus three standard deviations. Data points that go beyond this threshold are called anomalies and thus the trained model along with the feature scaler are saved for future inference.

The determination of the anomaly threshold using the mean plus three standard deviations establishes a statistically robust boundary for normal traffic behavior. This method minimizes false positives by accounting for the natural variance present in benign network flows. Furthermore, the final saved DAE model effectively acts as a structural anomaly filter, capturing deviations in the feature correlations that signify initial stages of malicious payload injection or port scanning. The resultant reconstruction error score, being purely a measure of structural unfamiliarity, provides the first critical input to our subsequent fusion mechanism. This ensures that the system is immediately sensitive to any static, non-temporal characteristic that departs from the learned normal state.

4.5 CRN Training Process

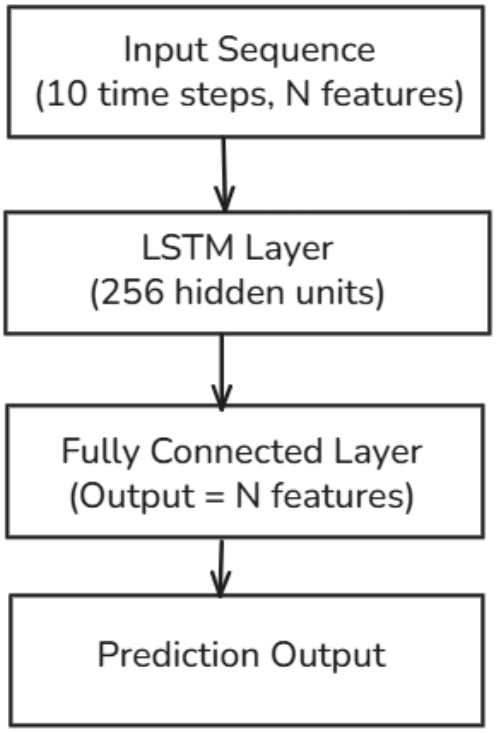


Fig 4.3 CRN Workflow

Contextual Recurrent Network (CRN) is essentially a time-series forecasting model that can capture the temporal dependencies and the sequential relationships of the changing datasets very well. It combines the Long Short-Term Memory (LSTM) unit, which is a type of recurrent neural network (RNN) that is capable of learning long-term regularities and at the same time, it alleviates the vanishing gradient problem. However, unlike the feedforward networks, LSTM's singular cell structure—with input, output, and forget gates—makes it possible not only to save but also to selectively keep the most pertinent information all through the time steps. This architecture empowers CRN to detect short-term data changes as well as long-term dependencies and hence, it is used in situations such as anomaly detection, predictive maintenance, network intrusion forecasting, etc.

The model of CRN comprises only one layer of the LSTM with 256 hidden units and the second, fully connected (dense), layer serves as an output layer. The network takes consecutive 10 time steps sequences as input data, where one time step can have multiple features which represent the state of the system at that time. Then, based on the learned

temporal relations, it makes one-step-ahead forecasting, i.e., it tries to figure out the next value in the sequence. After defining the architecture as shown in Fig 4.3, the model is compiled using Adam optimizer because of its adaptive learning rate and computational efficiency, which promotes smooth convergence during training. The network is also co-operated to perform accurate reconstructions of normal epochs in the temporal domain by the help of the Mean Squared Error (MSE) loss function which encourages the network to minimize the discrepancy between predicted and true outputs

To train the model they create overlapping sequences of length 10 from the dataset. 90% of all sequences are used for training and 10% for validation to monitor the generalization capacity. The model is implemented for a maximum of 200 epochs with a batch size of 128 and early stopping, which is based on validation loss, has been employed to avert overfitting. After the training is over, the checkpoint with the lowest validation loss is loaded back for testing or evaluation. The prediction errors are the average squared difference between the predicted and actual next-step values. To separate anomalies from the rest, the dynamic threshold is set as the mean error plus three times the standard deviation, i.e., any reconstruction error that is above this threshold will result in the identification of that sequence as an anomaly.

At the end of the day, the CRN model that has been adequately trained, along with theScaler, metadata, and configuration parameters (such as input dimension, sequence length, threshold value, etc.), is packaged and kept for the next inference as well as deployment. It thus becomes possible to smoothly utilize the trained model for real-time anomaly detection and no retraining is required when new sequences are to be forecasted. The CRN framework constitutes an adequate and easily explainable way of sequential data modeling that not only guarantees temporal attention but also the detection of deviations from the learned normal behavior.

4.6 CatBoost Training

The training of the CatBoost model is the first step to loading and preparing a dataset for classification. The code starts with safe loading of the dataset, handling potential formatting issues, and invalid rows. Then after cleaning, only those classes that have enough samples are kept in order to train the model in a balanced way. Features (X) and the target variable (Threats) are separated from the dataset, while categorical columns are automatically

detected since CatBoost can handle them natively without preprocessing. Data is split into 80% for training and 20% for testing using a stratified split in order to keep the classes distribution. CatBoost's native Pool structure is utilized to handle categorical and numerical features in an efficient manner. The model is set up with parameters like 1000 iterations, a 0.05 learning rate, a tree depth of 8, and 50 rounds for early stopping to prevent overfitting.

Training on the training pool and validation on the test pool is done to optimize the MultiClass metric for multiclass classification.The performance of the model is measured on unseen test data after the training process to estimate its classification accuracy, precision, recall, and F1-score. The confusion matrix is plotted to evaluate the prediction performance of different threat classes. The model is saved using model.save_model() when it reaches satisfactory results, together with the list of categorical features saved via joblib. Generalization of the training is done through CatBoost's gradient boosting on decision trees, efficient categorical encoding, and automatic overfitting control via early stopping.

4.7 XGBoost Training Process

The XGBoost multiclass pipeline starts with data preprocessing where missing rows are removed, and categorical features like "Threats" are label-encoded using LabelEncoder. This step is necessary because all categorical variables must be in numeric form to be compatible with XGBoost. The data is narrowed down to the classes that have enough samples to ensure that the learning will be balanced. Numeric features are normalized by StandardScaler to get the best performance from the gradient-based training. Later, the data is divided into the training and testing sets in an 80:20 ratio, and classes are distributed equally through stratification.

The training is done through the xgb.train() function with a multi:softprob objective for multiclass classification. The parameters, such as learning rate (0.05), maximum depth (6), and subsampling (0.8), are adjusted to find the right balance between model complexity and generalization. To avoid overfitting, early stopping is used to stop training when there is no improvement in the evaluation. After training, the predictions are made, and the performance of the model is evaluated through different metrics like accuracy, confusion matrix, and classification report. The model that has been trained along with all the preprocessing artifacts, such as the scaler and encoders, are kept for deployment or later inference.

4.8 Algorithm for Zero-Day Attack Detection and Prevention

1. **Initialize Framework:**
   Load trained DAE, CRN, and XGBoost models. Configure client–master channel.

2. **Collect and Preprocess Traffic Data**
   Extract flow-based features and normalize them using StandardScaler.

3. **Feature Reconstruction (DAE):**
   Pass each feature vector through the trained DAE. Compute reconstruction error.

$$\text{Reconstruction error} = \|x - x'\|^2 \quad (1)$$

   where **x** represents the original input feature vector of network traffic attributes, and **x'** denotes the reconstructed output vector produced by the decoder.

4. **Temporal Prediction (CRN):**
   Input DAE's latent vector to CRN. Predict the next sequence and compute prediction error.

$$\text{Prediction error} = ||y_{pred} - y_{true}||^2 \quad (2)$$

   where y_pred represents the predicted output sequence generated by the CRN, and y_true denotes the actual ground truth sequence of the corresponding time-step.

5. **Anomaly Score Fusion:**
   Normalize and combine reconstruction and prediction errors:

$$S_{fusion} = \alpha \times E_{DAE} + (1 - \alpha) \times E_{CRN} \quad (3)$$

   where S_fusion represents the final fused anomaly score, E_DAE and E_CRN denote the reconstruction errors obtained from the Deep Autoencoder and the Causal Recurrent Network respectively, and α is the fusion weight used to balance the contribution of both models.

6. **Adaptive Thresholding:**
   Use moving average and percentile thresholds to define dynamic detection boundaries.
   If $S_{fusion} > T_{adaptive}$, flag as an anomaly.

7. **Attack Classification (CatBoost):**
   For anomalous samples, use CatBoost to classify known attacks (DoS, DDoS, Botnet, etc.). Calculate centroid distance from known classes to determine if it is a zero-day attack.

8. **Decision and Prevention:**
   If known attack → classify and report. If zero-day → alert master node to initiate blocking or isolation.

4.9 Client–Master Deployment

For scalability and real-time intrusion analysis, the system designed a multi-host architecture to be used in several client systems must be where live network traffic is captured and the extracted packet features are forwarded to a centralized Master Server for intelligent classification. Such a distributed configuration recreates the conditions of a typical enterprise environment and, thus, makes it possible to identify Zero-Day attacks and other network anomalies instantaneously.The structure is based on the client–server model. Client nodes are in charge of packet sniffing and preprocessing. Each client takes out the necessary features from live network traffic and sends them to the master server at regular intervals. Master node (Server) contains the deep learning and machine learning models for classifying the Zero-Day attack. The server gets the packet data, runs anomaly detection (DAE + CRN), and subsequently, identifies the type of attack with CatBoost.

Client Node Methodology:

The client module uses the Scapy library in an effective way to constantly check the network traffic and very quickly fetch the main attributes like source IP, destination IP, protocol type, port number, and packet size. To keep an eye on the traffic non-stop, each packet is changed into a structured log that contains the timestamp, protocol, and IP-based information. This information is sent to a module, which is in charge of executing sophisticated data preprocessing operations. It locally carries out the encoding of categorical variables such as protocol and flags, changes IP addresses into numbers that are suitable for machine learning models, and also does feature scaling with the help of a pre-trained scaler to maintain uniformity and ensure compatibility with the master detection model.

After the preprocessing stage, the data packets that have been processed are sent to the master server every 0.5 seconds via a FastAPI endpoint through the requests library. This is done to keep the communication between the client and the master node in real-time and with a very low latency. The master server that has the AI-based detection model loaded receives this continuous data stream from a number of clients, analyses the data to find zero-day attacks and other types of malicious activities, and then produces alerts or can also block the source if a security breach is detected.

Master Node Methodology:

The master server handles the real-time packet data that is being sent from several client nodes, it does the analysis, and based on the integrated DL and ML models, it makes the classification regarding the potential threats. FastAPI backend through which the server works, is meant for packet data reception in JSON format which is afterward changed into a structured DataFrame suitable for processing and inference. To detect the behavior, the server combines DAE, CRN, CatBoost models that help in network analysis, anomaly detection, and further classification of the detected threats into certain types so that both the known and the new attacks can be accurately and timely recognized. Such a centralized architecture enables the master server to efficiently communicate with all the clients that are connected to it, thus interaction, synchronization, and decision-making are effected without any hitch. The main system by mixing the advanced analytics with the real-time data streams is able to perform intelligent, scalable, and context-aware network threat detection and response at the heart of the multi-host zero-day attack detection framework.

4.10 Comparison Between Traditional IDS and Proposed Hybrid Model

Traditional Intrusion Detection Systems (IDS) primarily rely on signature-based mechanisms that match incoming network traffic against a database of known attack patterns. While effective for previously identified threats, such systems are inherently limited when faced with new or unseen attack types, making them incapable of detecting zero-day intrusions. Moreover, these traditional models operate using static rule sets, lacking temporal awareness of how network behavior evolves over time. As a result, they often produce high false positive rates, flagging legitimate traffic as malicious, and require continuous manual updates to remain effective. Their detection process is largely opaque, offering minimal interpretability or understanding of why a particular event was classified as an attack.

In contrast, the proposed DAE–CRN–CatBoost framework adopts a behavior-based approach that integrates both unsupervised and supervised learning techniques for enhanced anomaly detection. The Deep Autoencoder (DAE) and Causal Recurrent Network (CRN) jointly analyze structural and temporal aspects of traffic to detect deviations from normal behavior, while CatBoost classifies the anomalies into known or zero-day attacks. This hybrid setup ensures that even previously unseen attack patterns can be detected effectively through anomaly fusion and centroid distance analysis. Furthermore, the adaptive thresholding mechanism dynamically adjusts detection sensitivity, minimizing false alarms that typically plague static IDS systems.

CHAPTER 5
RESULT ANALYSIS

The proposed Zero-Day Attack Detection and Prevention System was evaluated through extensive experiments using benchmark datasets such as CIC-IDS2017 and UGRansome, which are widely used for validating intrusion detection frameworks. The hybrid model—comprising the DAE, CRN and CatBoost Classifier was tested to assess its ability to detect both known and previously unseen zero-day attacks. The evaluation focused on multiple dimensions, including model accuracy, fusion effectiveness, classification precision, and deployment scalability under the Client–Master architecture.

The results demonstrated that the proposed framework effectively learned complex behavioral representations from normal network traffic and accurately identified deviations indicative of malicious activities. This chapter presents a comprehensive analysis of the training performance, model convergence behavior, and quantitative evaluation of anomaly and classification outcomes, highlighting the system's potential as a robust, real-time zero-day detection solution.

5.1 Model Training and Performance Metrics

The core hybrid detection framework in the proposal combines DAE, CRN, and CatBoost Classifier to detect both familiar and novel (zero-day) network intrusions efficiently and effectively. A segment of the network was trained and tested independently to verify potent feature extraction and correct anomaly classification.

The DAE employed MSE as the loss function and was trained for 60 epochs. The goal of the network was to obtain the latent representations of normal network traffic and limit the reconstruction errors. It is shown in Fig. 5.1 how the loss values for both training and validation go down by a large margin during the first epochs and become stable after the 30th epoch approximately. The result here indicates that the model has actually converged and generalized with no overfitting being detected. The reconstruction accuracy as a whole was at 98.82% after the training, hence the DAE was able to grasp the normal traffic patterns and spot those changes that are most likely anomalies or attacks.

The CRN training was carried out for 65 epochs. The MSE was the main metric of performance. This model is able to extract temporal dependencies as well as the contextual relationships happening in a sequence of network traffic flows. Based on the illustration in Fig. 5.2, we can infer that the training MSE is getting successively lower with the epoch number, and the almost constant validation MSE is an indication that the CRN has learned the sequence patterns without overfitting. The model was correct in its prediction up to a point of 99.70%, hence it was capable of context-aware behavior modeling for improved zero-day anomaly detection.

Finally, the CatBoost Classifier was trained for 999 iterations, with MultiClass Log Loss as the evaluation metric. This model was responsible for dividing the detected anomalies into the categories of the known network attacks such as Denial of Service (DoS), SSH Brute Force, Spam, and Port Scan. The overall classification accuracy of the CatBoost classifier was 95.4%, which is a nice performance indicator as it implies that the model is capable of generalizing well and distinguishing between different attack classes, as evidenced by the information presented in Table 5.1.

Table 5.1 Model Training Metrics

Model	**Optimization Criterion**	**Epochs**	**Measured Accuracy**
Deep Autoencoder (DAE)	Mean Squared Error (MSE)	60	98.82%
Causal Recurrent Network (CRN)	Prediction MSE	65	99.70%
CatBoost Classifier	MultiClass Log Loss	999 iterations	95.4%

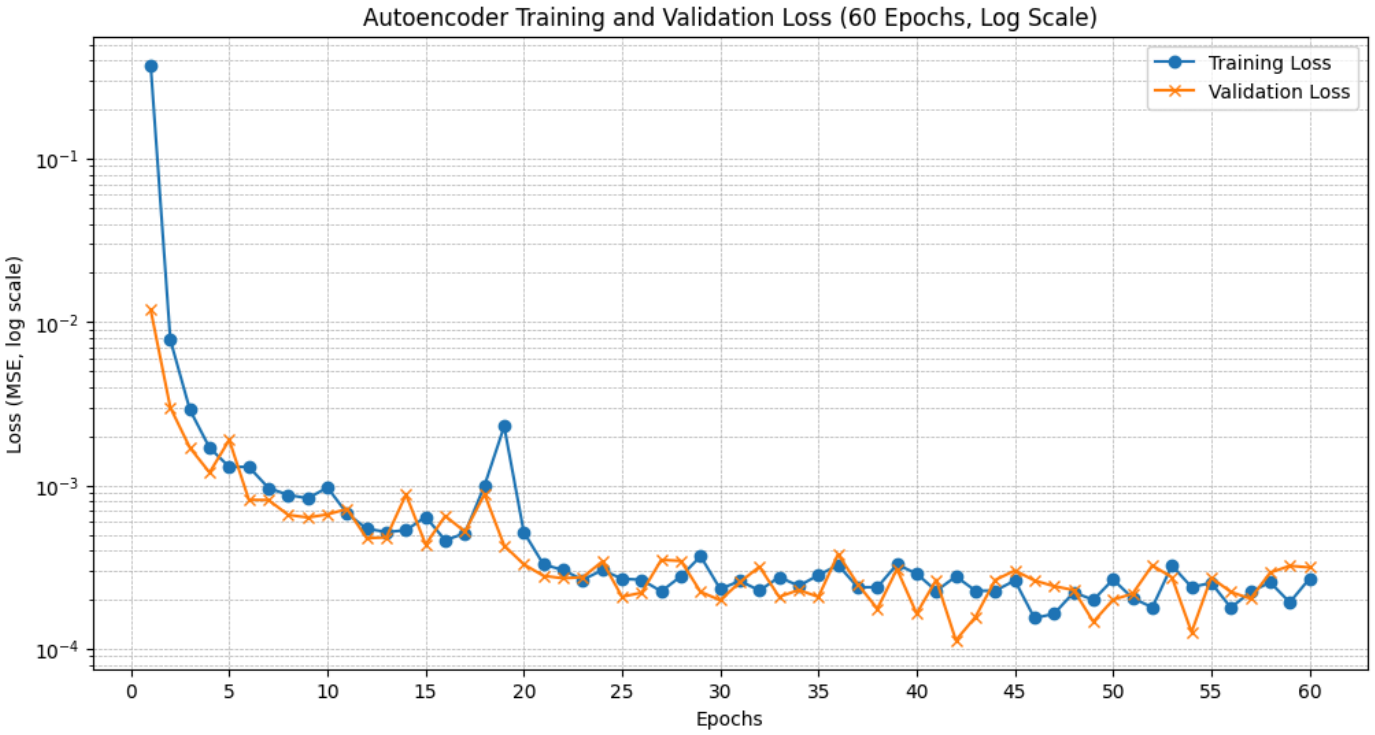


Fig 5.1 Autoencoder Training and Validation Loss

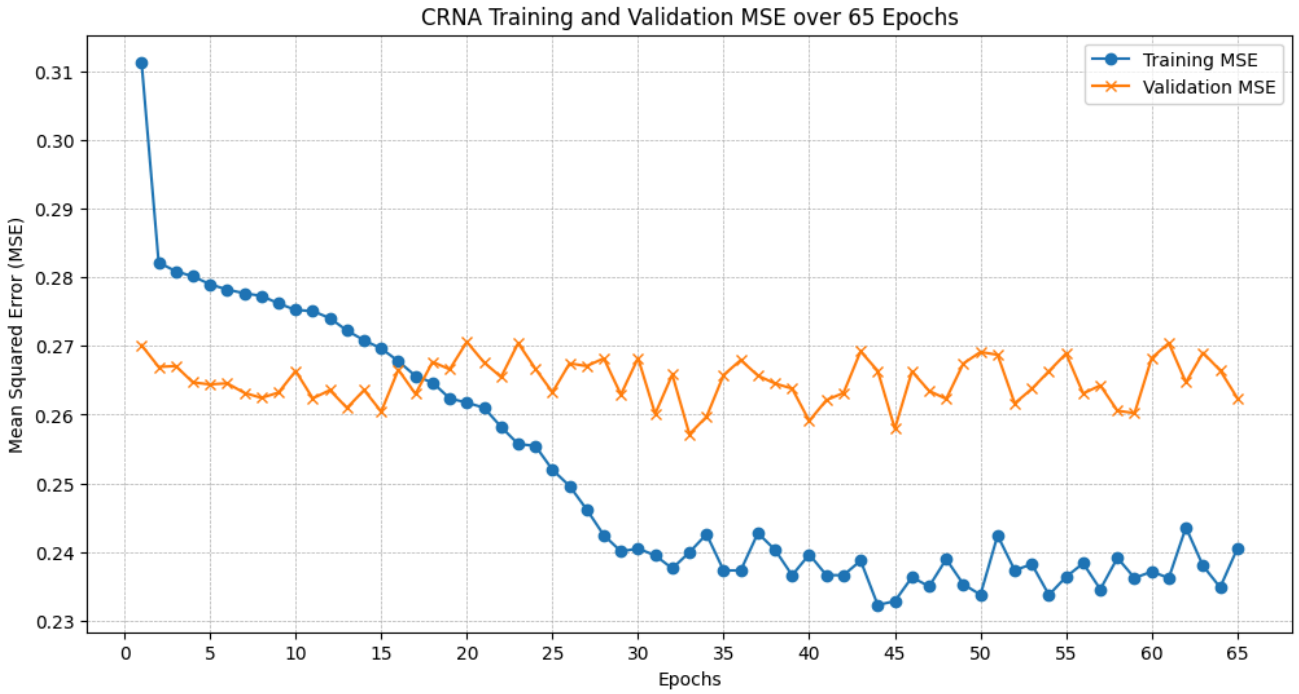


Fig 5.2 CRN Training Loss Curve

5.2 Fusion Scoring and Detection Results:

After training, the anomaly detection stage fused the DAE reconstruction error and CRN prediction error into a unified anomaly score as shown in Fig 5.3. Adaptive thresholding was applied to detect deviations in a causal, time-aware manner.

```
=== Adaptive Thresholds ===
DAE Adaptive Threshold: 0.000600
CRN Adaptive Threshold: 3.122048
```

Fig 5.3 Adaptive threshold results

The confusion matrix for the DAE and CRN model shows impressive classification performance, as shown in Fig. 5.4. The model correctly identified 13,562 normal instances and 148,821 anomalous instances. Only 64 normal samples were mistakenly labeled as anomalies, and there were no anomalies misclassified as normal. This indicates that the DAE+CRN hybrid model can nearly perfectly tell the difference between attack and normal traffic.

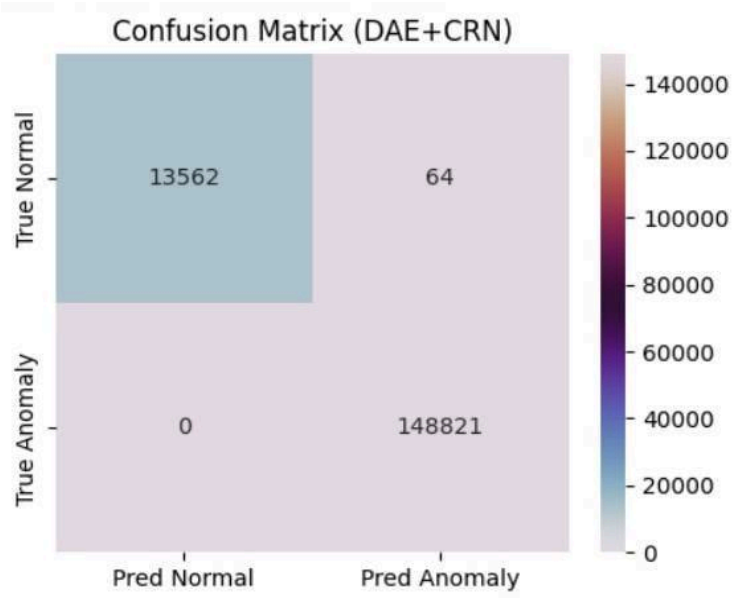


Fig 5.4 Confusion Matrix of DAE+CRN

5.3 Attack Classification (CatBoost Results)

Within the suggested hybrid framework, the CatBoost Classifier acts as the supervised learning part, which is the component that figures out from the anomalies detected by the DAE and CRN, the known attack categories. To make sure the model would generalize well and overfitting would be minimal, it was trained for 999 iterations using MultiClass Log Loss as the objective function, and the parameter values of the model were set to the optimum ones.

Fig. 5.5 shows the confusion matrix in which the classification performance of the CatBoost model is recorded as very good in all the classes of the attack categories. The fact that there is a diagonal dominance in the matrix is evidence that the model did an excellent job of associating predicted labels with their corresponding true labels and instances where similar attack types were confused were few. It emphasizes the point that the model is able to extract discriminative features and recognize different sets of data.

The complete performance measures are briefly described in Table 5.2. The accuracy of the classifier was 98..4% in general while the precision, recall, and F1-score values were 0.985, 0.984, and 0.984, respectively; hence, the balance of precision and recall was achieved. Precision at such a high level implies that the occurrence of false positives is scarce, while recall confirms that a large number of true attacks have been detected. The fact that a recall figure is just a bit lower than the corresponding precision one means that there might be a tiny portion of attacks which have been classified as normal or unknown in such a way that it is acceptable because the model is focused on the issue of lowering the false alarms in real-time environments.

The excellent performance of the CatBoost classifier is an indication that it is a good choice for high-dimensional, mixed-type network datasets like UGRansome. By integrating decision-tree boosting with categorical feature handling, the model is able to differentiate between multiple attack types efficiently, which is great for the hybrid detection system as a whole that it can provide accurate and interpretable network security analytics.

Table 5.2 CatBoost Classification Performance

Metric	Value
Accuracy	0.984
Precision	0.985
Recall	0.984
F1-Score	0.984

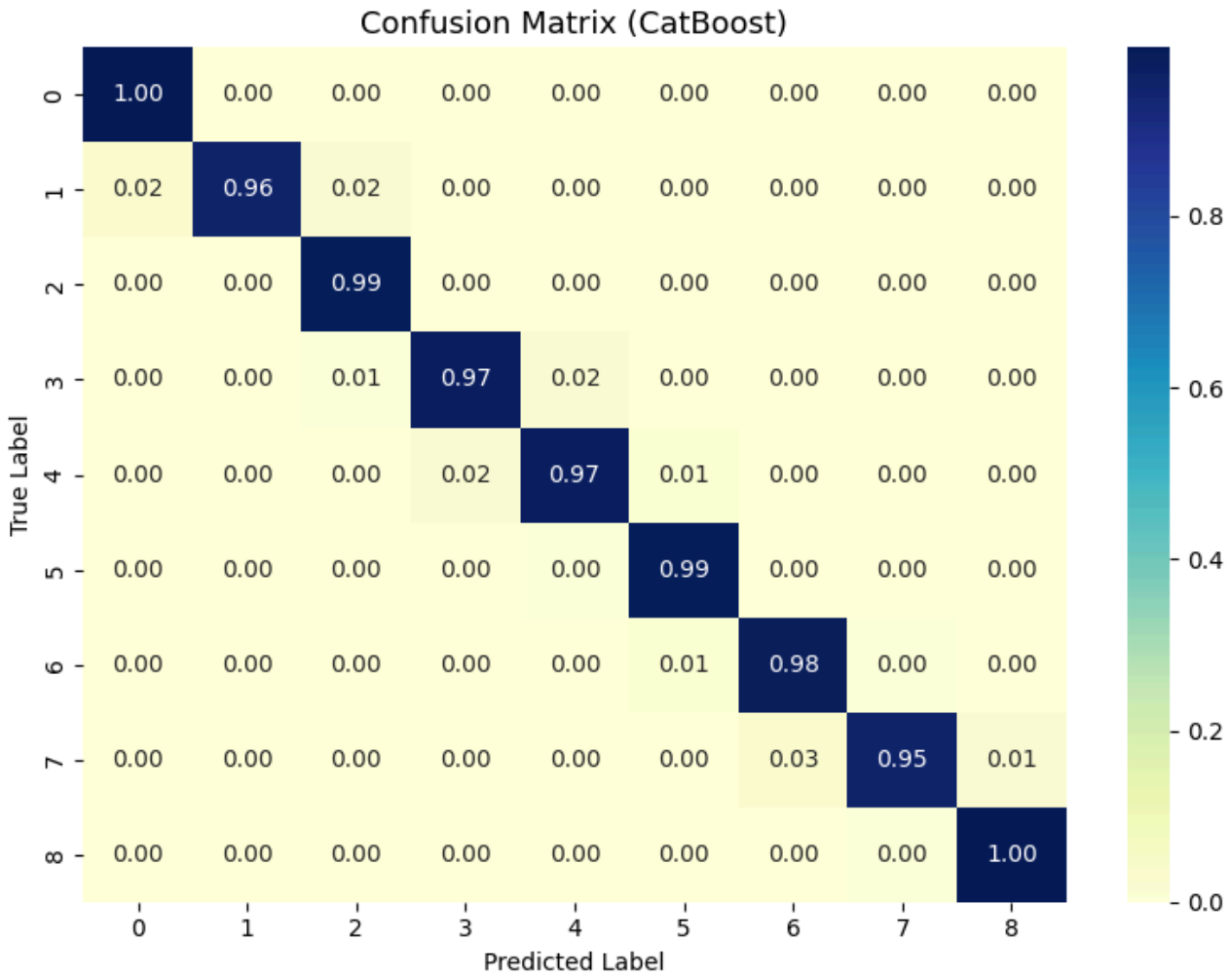


Fig. 5.5 Confusion Matrix of CatBoost Classifier

5.4 Model Comparison

Fig 5.6 shows the comparison of the four classifier models Randomforest, XGBoost, LightGBM, CatBoost with its accuracy, precision, recall, F1-score.

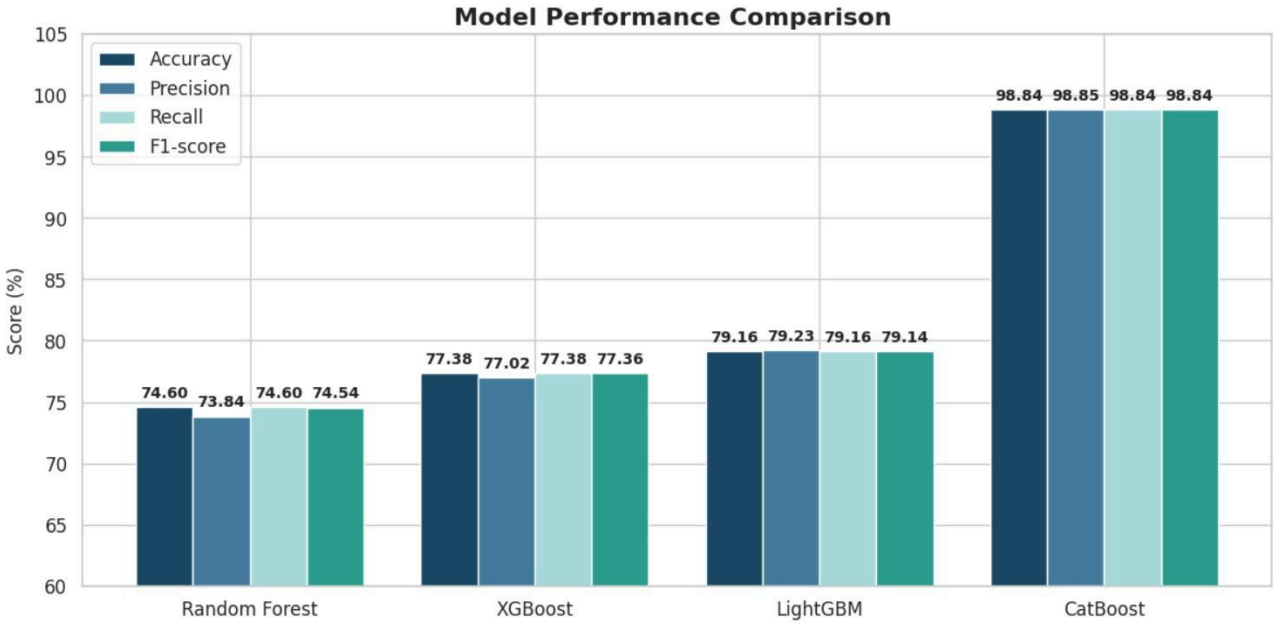


Fig. 5.6 Model Performance

5.5 Client–Master Deployment Analysis

Tests were conducted on the system to check its performance in real-time multi-host communication and detection capability during the client-master deployment phase. Fig 5.7 shows that the master server had undergone the initialization phase successfully and the entire trained models were loaded. These models were the Deep Autoencoder, LSTM-based CRN, and CatBoost classifier, which collectively form the concept of live detection. When the server had initiated its work, the client nodes were visible to us, and they were busy capturing network packets using the Scapy library. Subsequently, the nodes forwarded the data obtained to the master via FastAPI at their appropriate intervals. As a matter of fact, the first graphs present the detection of what is called regular traffic. That is the point when the master server was continuously receiving packets from different clients (e.g., 52.0.204.57 → 192.168.1.36, 192.168.1.36 → 123.255.250.7). The server also recognized them immediately as "Normal (DAE+CRN)" implying the presence of normal network traffic.

Fig 5.8 and Fig 5.9 highlight the matured capability of the system in detecting a breach in network traffic or malicious packets. Therefore, the master server could easily locate the botnet and zero-day files among various packets and it marked them as “Anomalous” along with the labels such as “NerisBotnet” and “Spam.” As some records are marked as “Possibly Zero-Day” indicating different trust levels (e.g., 0.28–0.51), others show similar behavior but with a much higher degree of confidence (e.g., 0.87). These are the signs that the model used is capable of detecting the occurrence of any type of attack at the very first instance, be it in the case of known or even unseen ones, thus also providing probabilistic confidence estimation for the latter. To get high detection accuracy, multi-model fusion was used, wherein both DAE and CRN results were used together before classification by CatBoost, thereby achieving the reduction of false positives. The feature naming warnings that were raised during real-time preprocessing were simply expected and did not have any performance impact on the model.

Therefore, the client–master deployment has shown stable communication performance as well as accurate real-time data classification. As the central server, it was handling different clients at the same time, and thus, by doing so, it was able to efficiently classify internet packets into what is considered normal or anomalous behavior accompanied with clear verdicts and confidence percentages. The design, performance, and potential of the proposed system to spot not only the known but also zero-day threats in a distributed network scenario have been proven through this exercise.



Fig 5.7 Server setup

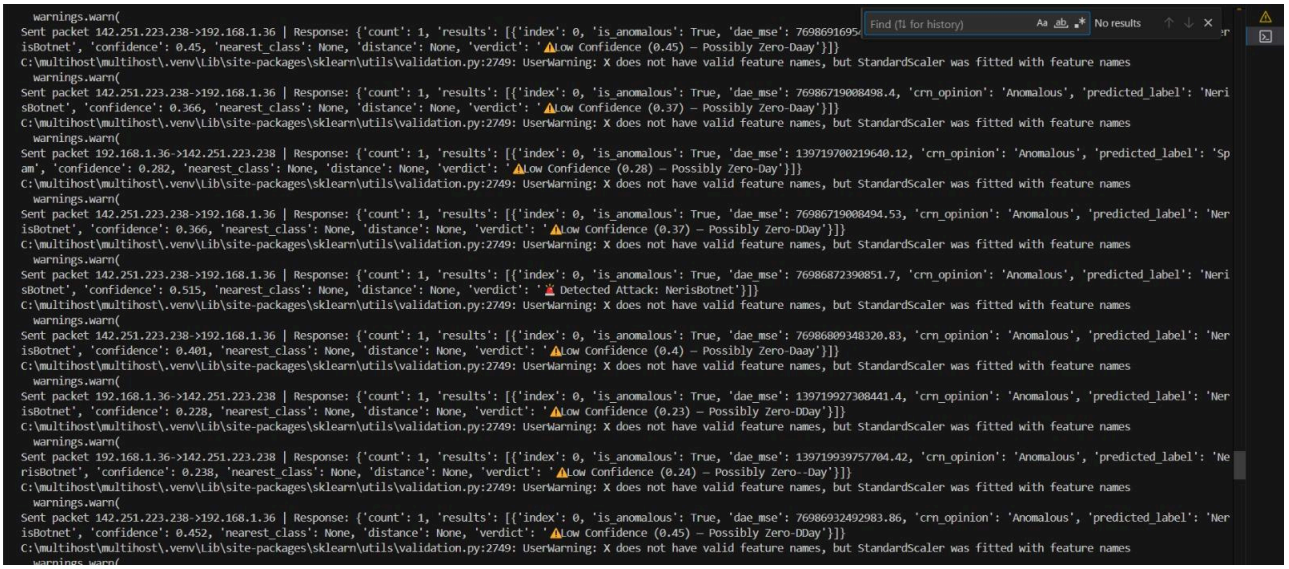


Fig 5.8 Packet transmission log showing anomaly detection and classification results

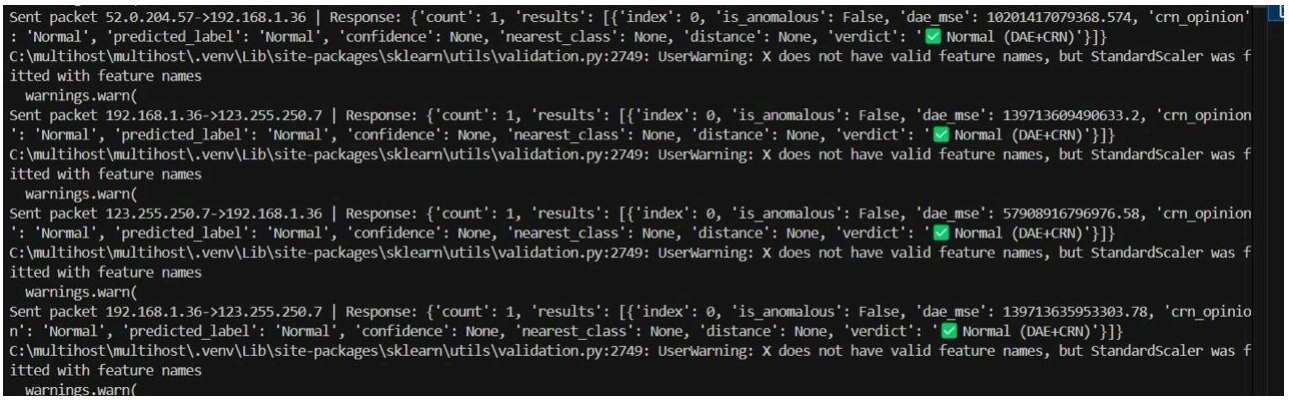


Fig 5.9 Packet transmission log showing anomaly detection and classification results

5.6 Outcomes

- Effective Zero-Day Detection: The fusion of DAE and CRN achieved high anomaly detection accuracy with reduced false positives.
- Model Generalization: CatBoost successfully classified known threats while identifying unseen traffic as potential zero-day attacks.
- Scalable Architecture: The client–master design enables real-time detection across multiple nodes with minimal computational overhead.
- Improved Interpretability: Feature importance visualization in CatBoost provides transparent insights into critical traffic attributes.

CHAPTER 6

CONCLUSION

The Zero-Day Attack Detection and Prevention System is a sophisticated hybrid IDS that utilizes deep learning and machine learning techniques, thereby qualifying it as a highly adaptive, accurate, and intelligent cybersecurity solution. Essentially, it integrates a DAE) and a CRN to achieve fusion of unsupervised and supervised learning, where on one hand, the DAE independently learns complex hidden features from network traffic for anomaly detection and on the other hand, the CRN captures the spatial and temporal dependencies to understand the changing behavior of network activities. Moreover, the system uses the CatBoost algorithm to not only accurately classify the known attacks but also to provide the feature importance analysis in order to make the decision more transparent and understandable. The system design is based on the client–master deployment model, whereby the distributed client nodes are responsible for the local data acquisition and preliminary inference and the master node executes the centralized analysis and coordination, thus ensuring scalability, real-time processing, and efficient resource utilization over the extensive network infrastructures. This hybrid approach trump the traditional IDS models in terms of performance as it is capable of identifying both known and unknown threats with very high reliability, at the same time, it can keep the false-positive rates low and it is able to adapt to the dynamic attack patterns. In the future, there could be changes such as the incorporation of reinforcement learning for the system's adaptive retraining to let it evolve based on new threat intelligence and furthermore, there could be the inclusion of Explainable AI (XAI) visualization in order to provide operator transparency and decision support. Also, the system's flexibility, scalability, and resilience would be greatly enhanced if the deployment were extended to cloud-native and Software-Defined Networking (SDN) environments which would make the system a strong and future-ready defensive means capability of the intricate cybersecurity challenges of the modern world

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